

	AMD Integrated Graphics Motherboards					
Brand	ASUS	ASUS	ASUS	Biostar	ECS	Gigabyte
Model No.	M4A78-HTPC/RC	M4A785TD-M EVO	M4A785TD-V EVO	TA785GE 128M	A785GM-M	MA785GM-US2H
Price (Rupees)	Rs. 7,100	Rs. 6,700	Rs. 6,700	Rs. 4,135	Rs. 4,800	Rs. 5,500
Overall (Out of 100)	56.26	60.58	67.21	48.90	60.53	55.85
Features (Out of 70)	31.39	35.16	40.55	22.89	35.20	31.60
Performance (Out of 30)	24.87	25.42	26.66	26.01	25.33	24.25
Features						
Chipset	AMD 780G	AMD 785G	AMD 785G	AMD 785G	AMD 785G	AMD 785G
Socket	AM3	AM3	AM3	AM3	AM3	AM3
Northbridge / Southbridge	780G / SB700	785G / SB710	785G / SB710	785G / SB710	785G / SB710	785G / SB710
Memory Type	DDR2	DDR3	DDR3	DDR2	DDR3	DDR2
Max Memory Supported (GB) / No of DIMMs	16 GB / 4	16 GB / 4	16 GB / 4	16 GB / 4	32 GB / 4	16 GB / 4
No. of SATA / IDE Ports	5 / 1	5 / 1	5 / 1	6 / 1	6 / 2	5 / 1
E-SATA Support / No of E-SATA Ports	Y / 1	Y / 1	Y / 1	N	Y / 2	Y / 1
No of PCI Express x16 graphics slots	1	1	2	1	1	1
No. of PCI / PCI x1 / PCI x4 Slots	1 / 2 / 0	2 / 1 / 0	3 / 1 / 0	2 / 0 / 0	1 / 2 / 0	2 / 1 / 0
Multi GPU (SLI/CrossFire/N) / No of PCIe Lanes	N / 0	N / 0	Y / 2	N / 0	N / 0	N / 0
Integrated Audio Controller	Y	VT1708S	VT1708S	Realtek ALC662	Realtek ALC 888S	Realtek ALC889A
Number of Sound channels	8	8	8	6	8	8
SPDIF / Optical Port	Y / Y	N / Y	N / Y	N / N	N / Y	N / Y
No of LAN Ports / 10/100 / Gigabit	1 / Gigabit	1 / Gigabit	1 / Gigabit	1 / Gigabit	1 / Gigabit	1 / Gigabit
Integrated Wi-Fi (Y/N)	N	N	N	N	N	N
No. of rear USB / FireWire Ports	4 / 0	6 / 1	6 / 1	4 / 0	6 / 1	6 / 1
All Solid State Caps (Y/N)	N	Y	Y	N	Y	Y
BIOS options						
CPU voltage range (volts)	N/A	1.125 ~ 1.7	1.125 ~ 1.7	N/A	N/A	0.825 ~ 2.025
CPU Voltage increments	0.05	0.00625	0.00625	0.05	N/A	0.025
Memory voltage range (volts)	1.8 ~ 2.1	1.1 ~ 1.61	1.1 ~ 1.61	1.95 ~ 2.65	N/A	N/A
Memory Voltage increments	0.1	0.015	0.015	0.1	0.01	0.1
Board Layout (So 10)						
Region around the CPU	8	8	8	8	7	8
Location of SATA ports	8	8	8	7	7	7
Placement of Memory modules	8	8	8	8	8	7
Graphics card slot and PCI slots	7	8	8	7	7	7
Package Bundle						
No of SATA / PATA Cables Provided	1 / 1	2 / 1	4 / 1	2 / 1	4 / 1	2 / 1
Mentionable Accessories	Remote control	-	-	-	-	-
Performance						
PC Mark 05						
CPU	10142	10278	10301	10220	10250	10293
Memory	6586	6365	6463	6307	6284	6578
Graphics	3372	2596	3738	3558	3417	2666
HDD	8719	8753	8761	8724	8754	8756
Overall	9341	7661	9391	9214	9235	7738
ScienceMark (Memory Bandwidth)	8242	10142	10168	9602	8609	8286
Cinebench R10	9203	11091	11142	11088	11057	11140
WinRAR 3.8	2146	2442	2401	2325	2267	2159
DivX Video Encoding (100 MB VOB) (sec)	31	30	30	31	31	31
Sequential File Transfer Test (8 GB File) (sec)	164	170	170	165	164	163
Random File Transfer Test (8 GB File) (sec)	195	187	185	191	191	190
Doom 3 FPS (800 x 600, medium detail)	55.3	60.1	60.3	57.1	55.3	55.5
3D Mark 2006 (CPU / Overall)	4618 / 1633	4691 / 1817	4698 / 1821	4655 / 1788	4682 / 1667	4652 / 1650
Company Of Heroes (1024 x 768, lowest)	40.9	43.4	43.5	40.8	34.6	39.6

PCIe x1 slot is too close to the heat sink and it looks like a close fit for any card. The two slots other than the PCIe x16 slot are the PCI slots. The Southbridge is just a big block of metal.

The MSI KA780G is a very basic looking board yet it packs in a lot of features. The 780G board comes with three PCI slots and a PCIe X1 slot above



Jetway PN78VM2-LF

the x16 slot for any sound cards. Any more PCIe x1 cards or x4 cards are out of the question. There's a basic overclocking switch positioned next to the Northbridge. There are also dedicated chunky buttons for the power, reset and clear CMOS functions.

The ECS A785GM-M is a really compact mATX board. The board looks

great with its dark gray heatsinks. The board does get pretty hot with the benchmarks running. The SATA ports on the sides are solid and divert the connectors away from the cards. There is only one PCI slot and two PCIe x1 slots one above and below the PCIe x16 slot. The heat sinks although close to the PCIe x1 slot but it's been cut out keeping in mind x1 cards. Being an